

AgriOps AI— Farm Task & Worker Management

Component 2 & Agent 2

Overview

- ❖ This handles day-to-day operations

```
Worker
WorkerSkill
FarmTask
TaskAssignment
TaskSchedule
TaskHistory
```

Component 2 handles the day-to-day operational execution of the AgriOps AI platform. While Component 1 establishes the structural and botanical layout of the farm (such as farms, fields, and crops), Component 2 manages the active labor, scheduling, and operational workflows required to keep the farm running smoothly.

For a standard four-person group, this component represents a complete vertical slice of the application, encompassing dedicated PostgreSQL database tables, Entity Framework Core configurations, C# ASP.NET Core REST API controllers, React administrative scheduling screens, and Flutter mobile views for field workers.

1. Database Entities and Relational Schema To support workforce management and task tracking, Component 2 uses a normalized relational design in PostgreSQL:

- **Worker:** Represents farm laborers, field operators, or specialized staff members.
Attributes: ***Id***, ***UserId*** (Foreign Key linking to system users), ***FullName***, ***ContactNumber***, ***EmploymentType***, ***Status***, ***CreatedAt***, ***UpdatedAt***.
- **WorkerSkill:** Maps specialized capabilities or certifications to specific workers (e.g., machinery operation, pesticide handling, irrigation management).
Attributes: ***Id***, ***WorkerId*** (Foreign Key), ***SkillName***, ***ProficiencyLevel***.
- **FarmTask:** Defines operational jobs that need to be performed on a field or crop season.
Attributes: ***Id***, ***FieldId*** (Foreign Key), ***CropSeasonId*** (Foreign Key), ***TaskType*** (e.g., Watering, Fertilization, Weeding), ***Priority***, ***Description***, ***TargetDate***, ***Status***, ***CreatedAt***, ***UpdatedAt***.
- **TaskAssignment:** Manages the multi-worker or single-worker allocation for a specific task.
Attributes: ***Id***, ***TaskId*** (Foreign Key), ***WorkerId*** (Foreign Key), ***AssignedDate***, ***Status***.
- **TaskSchedule:** Manages recurring or timed schedules for cyclical farm duties.
Attributes: ***Id***, ***TaskId*** (Foreign Key), ***Frequency***, ***StartDate***, ***EndDate***, ***NextExecutionDate***.
- **TaskHistory:** An audit trail tracking state transitions, execution logs, and evidence uploaded during task performance.
Attributes: ***Id***, ***TaskId*** (Foreign Key), ***PreviousStatus***, ***NewStatus***, ***ChangedByUserId***, ***Remarks***, ***EvidencePhotoUrl***, ***Timestamp***.

2. Key Features and Business Operations

Component 2 goes beyond simple CRUD operations by implementing structured business logic and operational status workflows:

➤ **Operational Task Orchestration:**

Enables managers or AI agents to schedule routine farm activities such as watering, fertilization, weeding, pest inspection, crop monitoring, harvesting, and equipment maintenance.

➤ **Strict Status Lifecycle Workflow:**

Replaces basic task updates with a controlled, multi-step operational pipeline:

1. AI or Farm Manager creates the task.
2. A worker is assigned.
3. The worker receives the task notification on their mobile app.
4. The worker performs the physical task in the field.
5. The worker uploads evidence (e.g., completion photos or notes) via Flutter.
6. The manager verifies the evidence on the React dashboard.
7. The task is marked as completed.

➤ **Workload Balancing and Skill Matching:**

Allow managers to assign tasks based on worker skills (e.g., ensuring only certified workers handle chemical fertilization tasks) and view current workload distributions to avoid overburdening specific staff.

➤ **Search, Filtering, and Sorting:**

The API supports advanced query parameters enabling users to filter tasks by status (e.g., Pending, In Progress, Verified), sort by priority or target date, and view history logs dynamically across clients.

3. Concrete Scenario Example

To visualize how Component 2 operates in a live environment:

➤ **Task Type:** Fertilization

➤ **Target Field:** Field A (Tomato - 2 acres)

➤ **Assigned Worker:** John Doe (Field Worker)

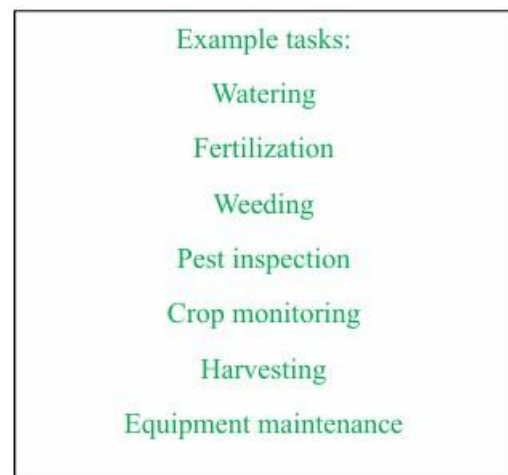
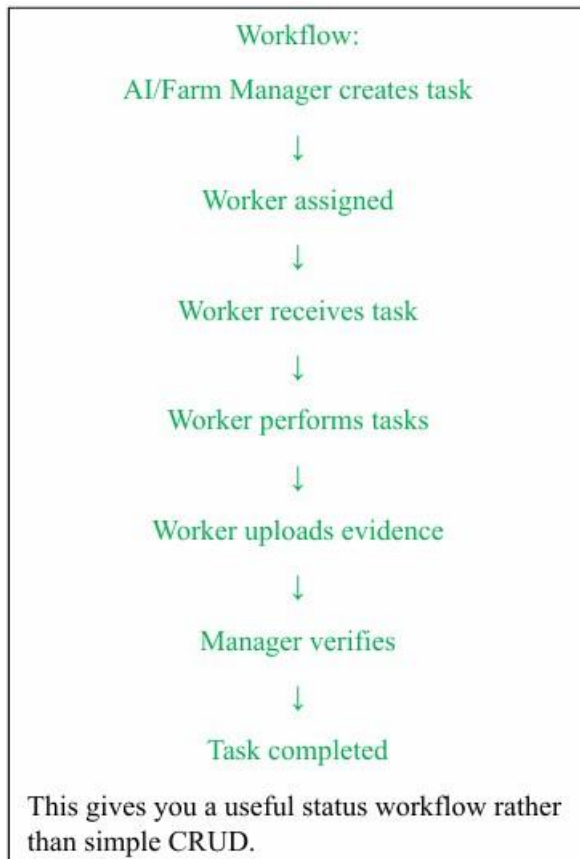
➤ **Workflow Execution:**

- Following an AI recommendation, the Farm Manager schedules a fertilizer application for Field A.
- John sees the task appear instantly on his Flutter mobile app dashboard under "Today's Tasks".
- John goes to Field A, performs the fertilization, and uses his mobile app's camera to upload an evidence photo of the applied field.
- The task status changes from In Progress to Pending Verification.
- The Farm Manager logs into the React web dashboard, reviews John's uploaded photo and notes, and clicks Verify/Approve, shifting the task status to Completed.

4. Integration with Other Components

Component 2 interacts closely with the rest of the application ecosystem:

- **Linked to Component 1 (Farm & Crop Management):** Every task must be anchored to a valid physical entity, such as a specific Field or CropSeason defined in Component 1.
- **Linked to Component 3 (Inventory & Resources):** When a task like Fertilization or Pest Inspection is executed, it can trigger inventory checks to deduct the exact amount of fertilizer or pesticide used from the stock repository.
- **Linked to Component 4 (AI Subsystem):** The AI Planning Agent and Weather Agents use Component 2 data to determine task timelines, rescheduling tasks automatically if weather conditions (like heavy rain) interfere with scheduled operations.



Agent 2 Overview- *Crop Analysis Agent*

Agent 2 (The Crop Analysis Agent) acts as the specialized visual and descriptive diagnostic support unit within the AgriMind AI multi-agent subsystem. Rather than operating as an isolated chatbot or a generic text generator, Agent 2 satisfies the SE3090 requirement for distinct, domain-relevant agents by fulfilling a precise role: parsing field observations and visual media submitted from the field to assess crop stress safely and systematically.

1. Role and Core Objectives

The Crop Analysis Agent is engineered to bridge the gap between raw, unstructured field data captured by field workers via mobile devices and actionable insights required by management.

Primary Responsibility: Ingest multimodal inputs (images and textual descriptions) to analyze plant health indicators and evaluate potential environmental or nutrient abnormalities.

Safety and Defense Architecture: To make the project practical and robust for a university evaluation, Agent 2 does not claim to definitively diagnose complex plant diseases or pathologies.

Instead, it provides an AI-assisted crop health assessment that highlights potential stress markers for human verification. This design decision makes the system significantly safer, more accurate, and much easier to defend during a technical viva.

2. Concrete Input Parameters

When a user initiates an assessment workflow from the field using the Flutter mobile application, Agent 2 consumes a structured input package containing:

- **Visual Data:** An uploaded crop image captured via the mobile device's camera (Crop image).
- **Descriptive Text:** A user-entered observation note, such as: "Leaves are becoming yellow."
- **Contextual Metadata:** Field ID, crop variety, and current growth stage (pulled from Component 1 tables).

3. Step-by-Step Execution Workflow

Upon ingestion, Agent 2 processes the data through a structured evaluation pipeline:

1. **Multimodal Ingestion:** Receives the image file and text string securely through the ASP.NET Core backend.
2. **Symptom Mapping & Classification:** Analyzes visual patterns (such as chlorosis or discoloration) alongside text inputs to map symptoms against known agronomic stress indicators.
3. **Risk & Priority Categorization:** Formulates a structured assessment output detailing potential issues, recommended actions, and priority levels.
4. **Handoff to Validation:** Forwards its classified output to Agent 4 (Validation & Safety Agent) and queues it for review by a qualified farm manager or agronomist on the React dashboard.

4. Output Contract Example

An example of the structured JSON contract produced by Agent 2 includes:

- **Potential issue:** Nutrient stress / environmental stress
- **Recommended action:** Perform field inspection
- **Priority:** Medium

5. Technical Implementation & Integration

- **Distinct Contract:** Agent 2 maintains an explicitly defined input/output contract separate from the planning and weather agents, satisfying the requirement for distinct agent responsibilities.
- **State Persistence:** Every analysis request, image reference, and generated assessment is logged in PostgreSQL under execution summary tables (AIWorkflows and AgentExecutions) to maintain a complete audit trail.